LGE Internal Use Only

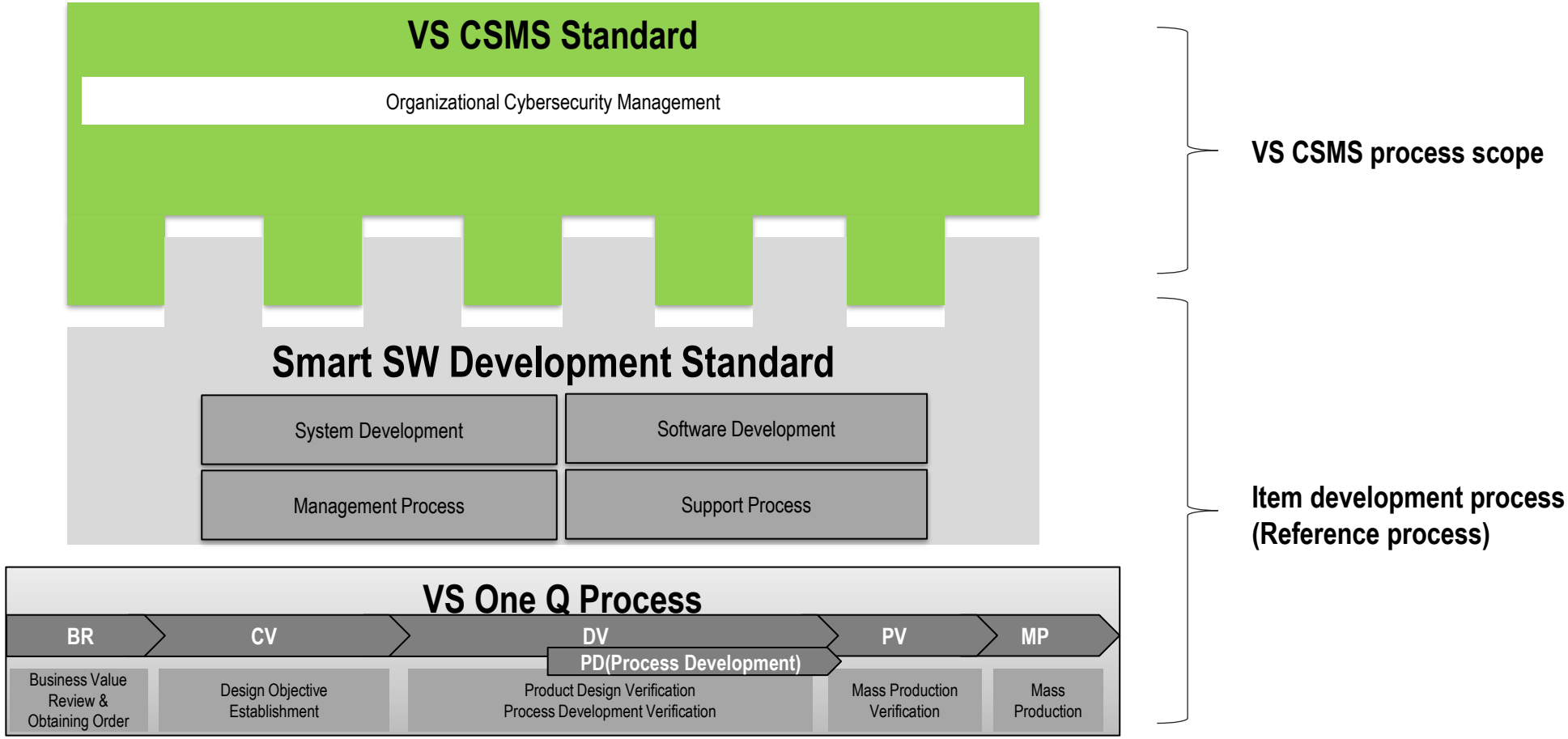
Understanding CSMS Standard – Production Processes

IVI LD1 – Cyber Security Part

March 5, 2026

1. VS CSMS standard composition

- The scope of the VS CSMS standard is limited to the cybersecurity area based on UNECE Cybersecurity Regulation, ISO/SAE 21434 and LGE Regulation of the SW Development Standard Process.
- To enable cybersecurity engineering, an organization institutes and maintains cybersecurity governance and cybersecurity culture. This involves specifying organization-specific rules and processes covering concept, development, production, operation, maintenance and decommissioning, including cybersecurity risk management, information sharing, vulnerability disclosure, cybersecurity monitoring, and incident response.
- Reference process applies the development process used by VS business unit.



2. Organization Structure of UNECE WP29

Organization structure

UN
*United Nations

UNECE
*United Nations Economic Commission For Europe

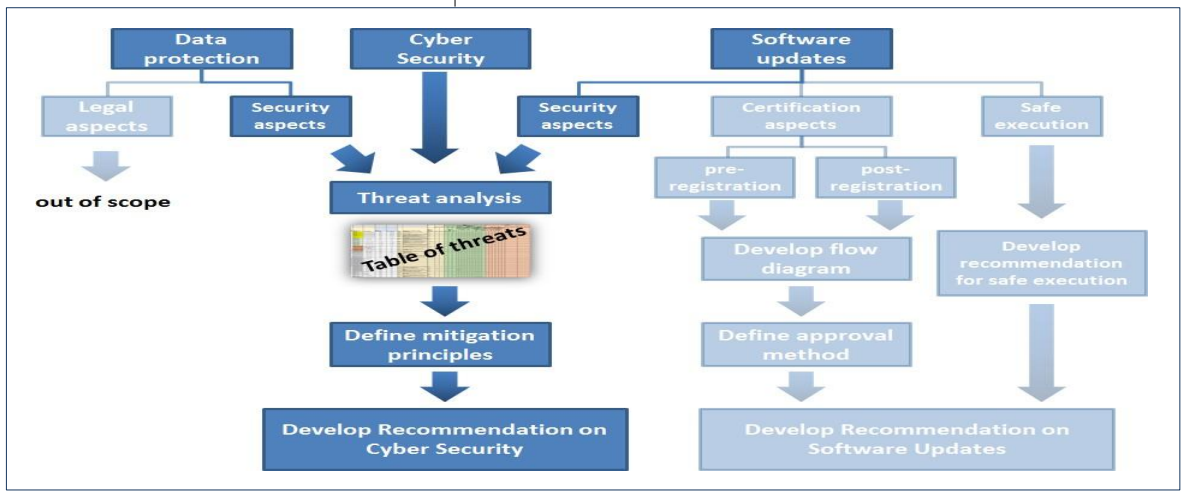
- UNECE : One of the five regional commissions under the jurisdiction of the United Nations Economic and Social Council. (56 members states)

ITC
*Inland Transport Committee

WP 29
*World Forum For Harmonization Of Vehicle Regulations

- WP29 : Unique worldwide regulatory forum within the institutional framework of the UNECE Inland Transport Committee.

GRVA
*Working Party on Automated/Autonomous and Connected Vehicles



- UNECE WP.29 enacted the Cyber Security Management System (CSMS).

CSMS
(Cybersecurity Management System)

3. UNECE WP29 R155 : Cybersecurity Regulation

Overview:

- As cybersecurity has emerged as an important criterion for automotive safety, the United Nations Economic Commission for Europe (UNECE) International Harmonization In June 2020, the UN Regulation No.155: Cybersecurity Regulation was adopted and came into effect on January 21.
- Applicable to new vehicles from July 2022 and all vehicles from July 2024 for countries subject to EU laws

Highlights:

❑ **Establish a CSMS (Cybersecurity Management System) and apply cybersecurity to vehicles in accordance with the CSMS.**

- [OEM] Acquisition of CSMS Certificate - > Acquisition of Vehicle VTA
- [Supplier] CSMS Implementation-> Support for OEM to acquire VTA (Evidence Provided)

❑ **Certificate of Compliance for CSMS (Process perspective)**

- Cybersecurity process certification for your cybersecurity organization, R&R, and vehicle lifecycle
- Design Perspective: Identify and manage cyber security risks
- Verify that identified risks are managed through testing, etc.
- Ongoing management of risk assessments
- Monitor and respond to cyberattacks
- Support analysis of all attacks (attempted or completed)
- Assess whether cybersecurity measures are effective against new threats and vulnerabilities

❑ **Vehicle Type Approval (Actual Vehicle Execution/Application Viewpoint)**

- Application of security technology by vehicle model and certification of evaluation results (actual vehicle, functional test)
- Evidence that you have a Cybersecurity Management System (CSMS) in place and that it has been applied to your vehicle.
- Analyze the results of the risk assessment and identify priorities (Critical)
- Mitigation to reduce cybersecurity risks
- Verification through cybersecurity testing to prove that countermeasures are working
- Action Points to Detect and Prevent Cyberattacks (IDPS)
- Security logger
- Appropriate monitoring for vehicle type and reporting monitoring activities to the approval body

OEM/Tier1 relations for UNECE WP29 R155 correspondence

OEM

- Establishment of CSMS system
- In accordance with the CSMS system Cybersecurity in vehicles application
- Acquired CSMS certification
- VTA (Vehicle Type Approval)

Supplier

- Establishment of CSMS system
- In accordance with the CSMS system Products in Cybersecurity application
- VTA acquisition point of OEM Evidence for the Circle offer

UNECE WP29 R155 and ISO/SAE 21434 Relationship

Regulation **UNECE WP29 R155** → Standard **ISO/SAE 21434**

❑ Requires CSMS at high level

- (7.2) Requirements for the Cyber Security Management System
- (7.2.1) For the assessment the Approval Authority or its Technical Service shall verify that the vehicle manufacturer has a Cyber Security Management System in place and shall verify its compliance with this Regulation.

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❑ Provides guideline how to implement a CSMS

- (Clause 5) Organizational cybersecurity management
- (Clause 6) Project dependent cybersecurity management
- (Clause 7) Distributed cybersecurity activities
- (Clause 8) Continuous cybersecurity activities

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- ISO/SAE 21434 supports to implement a CSMS as it is required by
- UNECE R155 ISO/SAE 21434 only describes the tasks for a process, but leaves the design to the users

4. CSMS activity definition

Organizational Cybersecurity Management

Organizational Cybersecurity Management

Cybersecurity Governance
Cybersecurity Culture
Information Sharing
Management System
Tool Management
Information Security Management
Organizational Cybersecurity Audit

Cybersecurity Concept Definition

Definition of Cybersecurity goals

Item Definition
Threat analysis and risk assessment(TARA)
Cybersecurity Goals specification

Definition of Cybersecurity requirements

Cybersecurity Requirements specification
Obtaining Cybersecurity requirements
Review Cybersecurity requirements

Cybersecurity System Development

Cybersecurity System design

Initiation of system development
(system)Cybersecurity requirements specification
Release of (system)cybersecurity requirements
System (cybersecurity) architectural design
Release of System (cybersecurity) architectural design
Cybersecurity System Integration test specification
Cybersecurity System Qualification test specification

Cybersecurity System Test & Validation

Cybersecurity System Integration Test
Cybersecurity System Qualification Test
(System) Penetration Test
Vehicle Validation
System Release

Cybersecurity HW Development

HW Cybersecurity Design

Initiation of HW development
(HW) cybersecurity requirements specification
Release of (HW) cybersecurity requirements
HW (cybersecurity) architecture design
HW (cybersecurity) design release
Cybersecurity HW Qualification Test Specification

HW Cybersecurity Verification

Cybersecurity HW Integration Test
Cybersecurity HW Qualification Test
HW release

Cybersecurity SW Development

SW Cybersecurity Design

Initiation of SW development
(SW)cybersecurity requirement specification
Release of (SW)cybersecurity requirement
SW (cybersecurity) architecture design
Release of SW (cybersecurity) architecture
SW unit cybersecurity design and implementation
Cybersecurity SW Qualification Test Specification

SW Cybersecurity Verification

Cybersecurity SW Unit Test
Cybersecurity SW Integration Test
Cybersecurity SW Qualification Test
Cybersecurity Vulnerability Test
SW Release

Cybersecurity Management & Supporting

Cybersecurity Management

Establish Release Cybersecurity Plan
Development of CIA & suppliers' CIA
Release of CIA of Cybersecurity Plan
Establish Cybersecurity Audit
Establish Cybersecurity Assessment

Cybersecurity Gate Review(Q-Gate)

Requirements confirm review
Feature Complete Review
Qualification completion Review

Supporting

Configuration Management
Requirement Change Management
Vulnerabilities Change Management

Production & Operation for Cybersecurity

Cybersecurity production

Production control plan

Continuous Cybersecurity Management

Cybersecurity threat monitoring
Initial incident analysis
Confirm affected products
Detailed threat analysis
Threat countermeasure development management
Prepare threat countermeasure
Post-Response Activities
End of Cybersecurity Support

Cybersecurity Case

Planning of Cybersecurity Case
Development of Cybersecurity Case

5. Steps in each product development stage

	DV	PV	Pre-MP	MP
LGE				
Objectives	Acquire Production Cybersecurity Requirements	Prepare cybersecurity documentations In compliance with requirements		Continue to update documentations from PV and perform assessment on Tier-2
CSM	**Product Design and Development** 1. Production Control Plan (http://collab.lge.com/main/x/rOdzrQ) 2. Production Cybersecurity Requirements (http://collab.lge.com/main/pages/viewpage.action?pageId=3184188188)	**Product and Process Validation** 3. Feasibility checking report of cybersecurity requirements. ([VSCSGU-4489] No.1 Process & Checklist - NEXT JIRA) 4. Applicability assessment report ([VSCSGU-4491] No.3 Process & Checklist - NEXT JIRA) 5. Archive assessment and inspection results ([VSCSGU-4492] No.4 Process & Checklist - NEXT JIRA) 6. SW Tools management process each project ([VSCSGU-4493] No.5 Process & Checklist - NEXT JIRA) 7. Security Tools Management category risk assessment report ([VSCSGU-4494] No.6 Process & Checklist - NEXT JIRA) 8. Manage cybersecurity competency tracking 9. Risk assessment of management processes (key management, transfer EDI, rework, … processes) [VSCSGU-4495] No.7 Process & Checklist - NEXT JIRA		**Mass Production** 10.Perform vulnerability monitoring and respond incident response activity ([VSCSGU-4500] No.12 Process & Checklist - NEXT JIRA) 11. Continue to collect assessment and inspection results 12. Follow buyer audit and prepare documents. 13.Conduct internal audit of each project 14. Monitor improvement and vulnerability remediation activities 15.Perform assessment on Tier-2 ([VSCSGU-4501] No.13 Process & Checklist - NEXT JIRA)
CSG	1. Provide regular training session ([VSCSGU-4498] No.10 Process & Checklist - NEXT JIRA) 2. Provide process training session ([VSCSGU-4495] No.7 Process & Checklist - NEXT JIRA) 3. Manage and archive education completion status & cybersecurity capability status of employees.([VSCSGU-4499] No.11 Process & Checklist - NEXT JIRA) 4. Establish and maintain processes including TARA, vulnerability management, incident response, vendor management for all projects 5. Collaborate with IT/security team to verify physical access security, information security and facility security ([VSCSGU-4496] No.8 Process & Checklist - NEXT JIRA) 6. Monitor IT information security certificates ([VSCSGU-4497] No.9 Process & Checklist - NEXT JIRA) 7. Archives assessment result and inspection results as evidence for VTA ([VSCSGU-4492] No.4 Process & Checklist - NEXT JIRA) 8. Audit preparation and reporting			

DV (Design Verification), IQ (Incoming Quality)
PV (Product Verification), OQ (Outgoing Quality)
TARA: Threat Analysis and Risk Assessment, VTA: Vehicle Type Approval

6. Main role in product development stage

No	Role	Phase	Template
1	Security part reviews production cybersecurity requirements and conduct feasibility checking whether the requirements can be applied and inspected in production line	PV	Feasibility ing report of cybe
2	Security part applies production cybersecurity requirements into production control plan and maintain the production control plan		
3	Security part performs assessment on inspection results to verify that the product is manufactured which is complying to production cybersecurity requirements in production control plan. (Ex. correct approved SW firmware/version shall be injected into product, correct keys shall be injected into product, removal of JTAG/Debug connector, secure boot enabled, sealed HW plug ,etc)		Applicability assessment report
4	Security part archives assessment result and inspection results as evidence for VTA		Archive nent and inspectio
5	Security part manages inventory of tools and equipment for (secure) production. (Ex. flashing software, test tools, ROM writing & Inspection tools such as Data management Software(DMS) / Tasklink / Flashcore / Lumenx FA & FCT Inspection Program)		SW Tools ment process each
6	Security part performs risk assessment regarding assets such as tools & equipment for (secure) production which shall be tool version management, obtaining rationale that tools are managed by strong access control and etc.		Security Tools agement category
7	Security part understands all process from start of production to end of production (OQC) and obtains evidence with rationale in perspective of cybersecurity of every step (ROM Writing ~ key injection ~ inspection ~ OQC ~ just before shipping)		Risk assessment nanagement proce
8	Security part collaborate with IT/security team to verify physical access security, information security and facility security.	All	CSG will implement
9	Security part monitor IT information security certificates such as TISAX, ISO 27001, IATF16949, etc		
10	Security part provides regular training session to employees, publish cybersecurity newsletter by translating to VietNam language of HQ newsletter and runs a cybersecurity campaign to enhance cybersecurity culture.		
11	Security part manages and archives education completion status & cybersecurity capability status of employees.		
12	Security part performs vulnerability monitoring on production line/system/tools and responds it as incident response activity.	MP	CSM continue to update documentations from PV and perform assessment on Tier-2
13	Security part performs assessment on Tier-2 (tool & equipment company, outsourced production company)		

Feasibility checking report of cybersecurity requirements.

Step	Input	Department / Team			Output	Lead time
		CSM	HQ ME	Relevant Teams (HQ SWPL, CSM & VH ME, PD, LD1)		
Acquire Document	•Cybersecurity requirements •Production control plan	1	Provide production control plan including production cybersecurity requirements		•Cybersecurity requirements •Production control plan	At the end of DV phase
Review Document	•Customer requirement •Production control plan •Internal requirement	2 Analysis requirements			Checklist items	D+3
Feasibility checking	Checklist items	3 With relevant teams feasibility checking OK Confirm cybersecurity requirements in production control plan	NG	Review & analysis regarding feasibility	Assessment report	D+5
Release final document	Assessment report	4		Release final production control plan	Production control plan	D+7

Applicability assessment report

Step	Input	Department / Team			Output	Lead time
		CSM	VH ME	Relevant Teams (VH PD, ETT, LD1)		
Acquire Document	Production control plan	1	Provide final production control plan including production cybersecurity requirements		HW, SW & tool requirements items	At final production control plan released
Review Document	HW, SW & tool requirements items	2 Determine execution teams			Checklist items	D+3
Feasibility checking	Checklist items	3 Applicability assessment (NG / OK)		Execute cybersecurity requirements into production process Provide inspection result	Assessment report	D+30 D+30 D+30
Release final document	Assessment report	4 Assessment report			Applicability assessment report	D+30

Archive assessment and inspection results

Step	Input	Department / Team		Output	Lead time
		CSM	CSG		
Acquire Document	• Assessment report • Inspection result	1 Provide cybersecurity assessment report and inspection result		• Assessment report • Inspection result	At applicability assessment report released
Review Document	• Assessment report • Inspection result	2	Collect documents	Summary file	D+3
Feasibility checking	Summary file	3 Review and update assessment report and inspection result	NG Accuracy assessment OK Confirm the accuracy of documents	Assessment report	D+7
Release final document	Assessment report	4	Document archiving	Evidence documents for VTA	D+7

SW Tools management process each project

Step	Input	Department / Team			Output	Lead time
		VH CSM	VH ME	Relevant Teams (VH ETT, PD, LD1)		
Acquire Document	• Production control plan • Tool versions list	1	Provide final production control plan including tool versions	Provide latest tool versions		At final production control plan released
Review Document	• Production control plan • Tool versions list	2 Collect tool version management file			Summary file	D+3
Feasibility checking	Summary file	3 Cross-check tool version information OK Confirm final tool versions	Update correct versions NG		Assessment report	D+5 D+5 D+5
Release final document	Assessment report	4 Archive final tool versions result			Tool version management report	D+5

Security Tools Management category risk assessment report

Step	Input	Department / Team		Output	Lead time
		VH CSM	Relevant Teams (VH ETT, PD, LD1)		
Acquire Document	• Tool version management report • Tools and equipment information	1	Provide identified assets such as tools and equipment information in production line		
Review Document	• Tool version management report • Tools and equipment information	2 Create a damage scenario		Damage scenario list	D+5
Analysis and assessment	Damage scenario list	3 Conduct impact rating based on S-F-O-P values Conduct threat scenario identification Attack path analysis Attack feasibility rating Risk value determination		• Impact rating result • Threat scenario list • Attack feasibility rating result	D+7
Release final document		4 Risk treatment decision		Threat analysis and risk assessment report	D+7

Risk assessment of management processes

Step	Input	Department / Team			Output	Lead time
		VH CSM	VH ME	Relevant Teams (VH ETT, PD, LD1, etc)		
Acquire Document	• Production control plan • Production route chart	1	Provide production processes in production control plan	Provide processes map	• Processes map	
Review Document	• Processes map	2 Analyze and identify processes for cybersecurity compliance and implementation			• Evidence list	D+5
Analysis and assessment	• Evidence list	3 Collect evidence of each process Conduct completeness and accuracy assessment Enough Confirm evidence of each process	Review and update missing content Not enough		• Evidence assessment result	D+5
Release final document		4 Archive evidence of each process			Evidence of each process report	D+5

LGE Internal Use Only

Thank you